

Verbal Autopsy with Participatory Action Research (VAPAR): Expanding the knowledge base through partnerships for action on health equity

VAPAR Evidence Review Lack of clean, safe water

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Water is a global issue

In rural communities around the world, lack of clean water, inadequate sanitation and poor hygiene are major contributors to avoidable death and disease. Lack of clean water increases vulnerability to conditions including diarrhoea, malnutrition, malaria, lymphatic filariasis, intestinal nematode infections, trachoma and schistosomiasis.(Prüss-Üstün et al., 2008) In 2012, 742,000 diarrhoea-related deaths were caused by lack of access to water and sanitation.(Prüss-Ustün et al., 2014) The impact of diarrhoea is most acute in children under 5, and is a leading cause of avoidable mortality in this age group.(UNICEF, 2017)

The importance of water and sanitation was emphasised in the Millennium Development Goals (MDGs) 1990-2015 during which time access expanded considerably. The target for safe drinking water was achieved in 2010 ahead of the 2015 deadline with 91% of the world population accessing safe drinking water, compared to 76% in 1990.(SSA, 2015a) An estimated 663 million people still lacked access in 2015, however, and the target on improved sanitation was not met.(UNICEF and WHO, 2015) The Sustainable Development Goals (SDGs) 2015-2030 commit to the unfinished water agenda, aiming to ensure available and sustainable management of water and sanitation for all.(UN, 2018)

Water is a human rights issue

The human and societal costs are extensive. In addition to the burden of avoidable mortality and morbidity outlined above, studies have identified increased risks of physical disability due to water carrying, a burden borne disproportionately by women and children.(Bisung and Elliott, 2017; Geere et al., 2018) Shame and emotional distress have been related to lack of water and that individual distress can erode social cohesion and capacity for community participation.(Bulled, 2017; Harris et al., 2018) Service delivery protests have also increased drastically and been associated with a combination of violent masculinities and a crisis of community representation in local government.(Chikulo, 2016; Langa and Kiguwa, 2013; Twala, 2014)

South Africa and the Right to Water

The government enshrined water as a basic human right in the 1996 constitution, 14 years before the 2010 UN declaration.(RSA, 1996; UNGA, 2010) The National Water Act of 1998, and the Water Services Act of 1997 were among a series of legislative and policy shifts to redress discrimination, promote equitable access and support municipalities to provide services, and in which cooperative governance and public participation were centralised.(Beck et al., 2016; RSA, 1997, 1998) Municipalities assumed statutory obligations to provide 6000 litres per household per month free of charge, and between 1994 and 2004 there was 15billion ZAR (3billion USD) of government investment in water and sanitation.(van Koppen and Schreiner, 2014) From 1990 to 2015, access increased 98% to 100% and 66% to 81% in urban and rural areas and the MDG targets were met.(SSA, 2015b)

Devolved responsibility?

While initially driven centrally, service provision and water supply were devolved to local government and municipalities in 2003 and 2006 respectively.(van Koppen and Schreiner, 2014; WSP, 2011) Ambitious decentralisation and developmental agendas relocated a range of responsibilities to local levels, where multiple constraints were faced. These included: financial distress, debt, inability to raise revenue, serious administrative and financial mis-management, neo-patrimonialism, tendering corruption and manipulation of public procurement.(Beresford, 2015; Van Der Mescht and Van Jaarsveld, 2012)

Municipalities were also unable to account for large amounts of complex technical information, with no household usage monitoring systems in most rural areas.(Kanyane, 2014)

Community-based water management was assigned to Catchment Management Agencies (CMAs) promoting public participation and Water User Associations (WUAs) as user co-operatives.(Karodia and Weston, 2001) There were challenges with capacity and clarity, however, with overlapping and ill-defined mandates, and links to community-based structures were limited.(Denby et al., n.d.) While CMAs and WUAs were “*meant to increase participation of stakeholders including communities in the management of water resource...efforts have not translated into effective participation ...there is no link between the national water quality management frameworks and community-based development structures*”.(Nare et al., 2011) Of the 19 CMAs established nationally, only two were operational in 2015.(Movik et al., 2016)

As well as advancing nation-building, the government embraced the international Integrated Water Resource Management (IWRM) paradigm (Movik et al., 2016). Over time, however, IWRM came to be seen as overly complex and technocratic, limiting the state’s role, with insufficient attention to context and integration, and overlooking equity and poverty alleviation.(Volenzo and Odiyo, 2018) This was followed by calls for a renewed focus on context in policy, strategic intervention and resource mobilisation with community participation as a unifying concept.(Casazza et al., 2016; Merrey, 2008) The second National Water Resource Strategy (NWRS2) acknowledges the role of water in social and economic development and commits to infrastructure, services and equity as a policy goals.(DWA, 2013)

Infrastructure problems

Despite visionary policy and legislation and considerable government investment in infrastructure, maintenance backlogs became a systemic challenge. In 2012 costs for outstanding maintenance reached USD 1.4billion and there are high proportions of households (78% in Mpumalanga and 70% in Limpopo) without basic services and interrupted supply due to non-functioning infrastructure, poor maintenance, and empty or insufficient water in reservoirs.(Sershen et al., 2016; van Koppen and Schreiner, 2014) Today around five million South Africans living mainly in rural areas do not have reliable access to drinking water.(SSA, 2016)

Adapted from (Hove et al., 2019)

REFERENCES

- Beck T, Rodina L, Luker E, et al. (2016) *PROGRAM ON WATER GOVERNANCE REPORT: INSTITUTIONAL AND POLICY MAPPING OF THE WATER SECTOR IN SOUTH AFRICA OVERVIEW The urban water landscape of South Africa INSTITUTIONAL AND POLICY MAPPING OF THE WATER SECTOR IN SOUTH AFRICA*. Available at: www.edges.ubc.ca (accessed 3 April 2019).
- Beresford A (2015) Power, patronage, and gatekeeper politics in South Africa. *African Affairs* 114(455). Narnia: 226–248. DOI: 10.1093/afraf/adu083.
- Bisung E and Elliott SJ (2017) Psychosocial impacts of the lack of access to water and sanitation in low- and middle-income countries: a scoping review. *Journal of Water and Health* 15(1): 17–30. DOI: 10.2166/wh.2016.158.
- Bulld N (2017) The Effects of Water Insecurity and Emotional Distress on Civic Action for Improved Water Infrastructure in Rural South Africa. *Medical Anthropology Quarterly* 31(1): 133–154. DOI: 10.1111/maq.12270.
- Casazza A, Regional U and Advisor S (2016) “Aligning the Sustainable Development Goals (SDGs) to the National Development Plan (NDP): Towards domestication of the SDGs in South Africa” Available at: http://www.statssa.gov.za/MDG/MDG_Country_Report_Final30Sep2015.pdf (accessed 17 December 2018).
- Chikulo BC (2016) “The Smoke That Calls”: A Review of Service Delivery Protests in South Africa 2005 -2014. *Journal of Human Ecology* 55(1–2). Routledge: 51–60. DOI: 10.1080/09709274.2016.11907009.
- Denby K, Movik S, Mehta L, et al. (n.d.) *The ‘Trickle Down’ of IWRM: A Case Study of Local-Level Realities in the Inkomati Water Management Area, South Africa*. Available at: www.water-alternatives.org (accessed 5 April 2019).
- DWA (2013) *National Water Resource Strategy: water for an equitable and sustainable future*. Pretoria. Available at: http://www.dwa.gov.za/documents/Other/Strategic_Plan/NWRS2-Final-email-version.pdf (accessed 11 December 2018).
- Geere J-A, Bartram J, Bates L, et al. (2018) Carrying water may be a major contributor to disability from musculoskeletal disorders in low income countries: a cross-sectional survey in South Africa, Ghana and Vietnam. *Journal of Global Health* 8(1): 010406. DOI: 10.7189/jogh.08.010406.
- Harris LM, Kleiber D, Rodina L, et al. (2018) Water Materialities and Citizen Engagement: Testing the Implications of Water Access and Quality for Community Engagement in Ghana and South Africa. *Society & Natural Resources* 31(1). Routledge: 89–105. DOI: 10.1080/08941920.2017.1364818.
- Hove J, D’Ambruoso L, Mabetha D, et al. (2019) ‘Water is life’: Developing community participation for clean water in rural South Africa. *BMJ Global Health* 4(3): 1–13. DOI: 10.1136/bmjgh-2018-001377.
- Kanyane M (2014) Exploring Challenges of Municipal Service Delivery in South Africa (1994 - 2013). 2014 2(1). 2014th-03–01 ed.: 21. DOI: 10.4102/apsdpr.v2i1.45.

- Karodia H and Weston DR (2001) *Intersectoral Management of River Basins South Africa's New Water Policy and Law*. Available at: <http://publications.iwmi.org/pdf/H029111.pdf> (accessed 3 April 2019).
- Langa M and Kiguwa P (2013) Violent masculinities and service delivery protests in post-apartheid South Africa: A case study of two communities in Mpumalanga. *Agenda* 27(1): 20–31. DOI: 10.1080/10130950.2013.793897.
- Merrey DJ (2008) Is normative integrated water resources management implementable? Charting a practical course with lessons from Southern Africa. *Physics and Chemistry of the Earth, Parts A/B/C* 33(8–13). Pergamon: 899–905. DOI: 10.1016/J.PCE.2008.06.026.
- Movik S, Mehta L, Van Koppen B, et al. (2016) Emergence, Interpretations and Translations of IWRM in South Africa. *Water Alternatives* 9(3): 456–72. Available at: www.water-alternatives.org (accessed 11 December 2018).
- Nare L, Odiyo JO, Francis J, et al. (2011) Framework for effective community participation in water quality management in Luvuvhu Catchment of South Africa. *Physics and Chemistry of the Earth, Parts A/B/C* 36(14–15). Pergamon: 1063–1070. DOI: 10.1016/J.PCE.2011.08.006.
- Prüss-Ustün A, Bartram J, Clasen T, et al. (2014) Burden of disease from inadequate water, sanitation and hygiene in low- and middle-income settings: a retrospective analysis of data from 145 countries. *Tropical Medicine & International Health* 19(8). Oxford, UK: BlackWell Publishing Ltd: 894–905. DOI: 10.1111/tmi.12329.
- Prüss-Ustün A, Bos R, Gore F, et al. (2008) *Safer water, better health : costs, benefits and sustainability of interventions to protect and promote health*. Geneva. Available at: https://apps.who.int/iris/bitstream/handle/10665/43840/9789241596435_eng.pdf?sequence=1&isAllowed=y (accessed 5 April 2019).
- RSA (1996) Constitution of the Republic of South Africa 1996. Republic of South Africa. Available at: <https://www.gov.za/documents/constitution-republic-south-africa-1996> (accessed 5 April 2019).
- RSA (1997) Water Services Act 1997. Act No. 108, 1997. Republic of South Africa. Available at: http://portal.unesco.org/en/files/47386/12670896091Water_Services_Act_1997.pdf/Water%2BServices%2BAct%2B1997.pdf (accessed 5 April 2019).
- RSA (1998) National Water Act 1998. Act No 36 of 1998. Republic of South Africa. Available at: http://www.dwa.gov.za/Documents/Legislature/nw_act/NWA.pdf (accessed 5 April 2019).
- Seršen, Rodda N, Stenström TA, et al. (2016) Water security in South Africa: Perceptions on public expectations and municipal obligations, governance and water re-use. *Water SA* 42: 456–465. Available at: http://www.scielo.org.za/scielo.php?script=sci_arttext&pid=S1816-79502016000300011&nrm=iso.
- SSA (2015a) *Millennium Development Goals 7: Ensure environmental sustainability*. Pretoria. Available at: http://www.statssa.gov.za/MDG/MDG_Goal7_report_2015_.pdf (accessed 17 December 2018).
- SSA (2015b) *Millennium Development Goals Country Report 2015*. Pretoria. Available at: http://www.statssa.gov.za/MDG/MDG_Country_Report_Final30Sep2015.pdf (accessed 17 December 2018).
- SSA (2016) *GHS Series Volume VIII Water and Sanitation: in-depth analysis of the General Household Survey 2002–2015 and Community Survey 2016 data*. Pretoria. Available at: www.statssa.gov.za (accessed 5 April 2019).
- Twala C (2014) The Causes and Socio-political Impact of the Service Delivery Protests to the South African Citizenry: A Real Public Discourse. *Journal of Social Sciences* 39(2). Routledge: 159–167. DOI: 10.1080/09718923.2014.11893279.
- UN (2018) *Sustainable Development Goal 6 synthesis report on water and sanitation 2018*. New York. Available at: https://sustainabledevelopment.un.org/content/documents/19901SDG6_SR2018_web_3.pdf (accessed 5 April 2019).
- UNGA (2010) 64/292. *The human right to water and sanitation*. New York. Available at: https://www.un.org/en/ga/search/view_doc.asp?symbol=A/RES/64/292.
- UNICEF (2017) *Thirsting for a Future Water and children in a changing climate*. New York. Available at: www.unicef.org/wash/waterandclimate (accessed 5 April 2019).
- UNICEF and WHO (2015) *Progress on sanitation and drinking water: 2015 update and MDG assessment*. New York/Geneva. Available at: www.wssinfo.org (accessed 5 April 2019).
- Van Der Mescht J and Van Jaarsveld M (2012) *Addressing operations and maintenance challenges in smaller municipalities*. Port Elizabeth. Available at: <http://www.infrastructurene.ws/wp-content/uploads/sites/4/2016/04/Addressing-operations-and-maintenance-challenges-in-smaller-municipalities-Johan-van-der-Mescht-Vela-VKE.pdf> (accessed 5 April 2019).
- van Koppen B and Schreiner B (2014) Moving beyond integrated water resource management: developmental water management in South Africa. *International Journal of Water Resources Development* 30(3): 543–558. DOI: 10.1080/07900627.2014.912111.
- Volenzo T and Odiyo J (2018) Ecological Public Health and Participatory Planning and Assessment Dilemmas: The Case of Water Resources Management. *International Journal of Environmental Research and Public Health* 15(8): 1635. DOI: 10.3390/ijerph15081635.
- WSP (2011) *Water supply and sanitation in South Africa: turning finance into services for 2015 and beyond*. Available at: www.amcow.net (accessed 5 April 2019).

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